

VAIBHAV.P.K

vaibhavpk52@gmail.com | +91-8148949904 | linkedin.com/in/vaibhav-pk | github.com/Devvaibhavpk

Summary

Computer Science undergraduate specializing in front-end development, data science, and machine learning model building. Experienced in developing responsive web applications using the MERN stack and implementing AI-driven solutions through academic projects. Skilled in translating data into insights and building efficient user-focused applications. Passionate about leveraging technology to create impactful and scalable solutions.

Work Experience

Research Intern – Indira Gandhi Centre for Atomic Research (IGCAR)

June 2025 – July 2025

- Collected and analyzed soil samples from multiple districts to detect radioactive traces of thorium, radium, and uranium.
- Measured and evaluated dosage rates of radioactive elements for environmental safety assessment.
- Built a prediction model using K-Nearest Neighbors (KNN) to forecast radioactive concentration levels.
- Gained practical experience in data preprocessing, model building, and scientific research methodologies.

Event Operations and Management Lead, Open Source Programming Club, VITC

- Organized 4+ national-level 24-hour hackathons, coordinating end-to-end event management, logistics, and volunteer teams to ensure smooth execution and participant engagement.

Education

Vellore Institute of Technology, Chennai

Aug 2023 – Aug 2027

B.Tech, CSE (Data Science)

Sri Sankara Vidyalaya, Mat, Hr, Sec School

June 2012 – May 2022

Skills

Languages	C++, Java, Python, JavaScript, TypeScript, SQL, HTML, CSS
Tools/Frameworks	React, Node.js, Express.js, MongoDB, Clerk, scikit-learn, Pandas, NumPy, Git
Other Skills/Interests	Leadership, Team Management, Guitarist, Fitness & Strength Training
Languages Known	English, Hindi, Marathi, Tamil, Telugu

Projects

Agriedge

- IoT-enabled platform using React, Node.js, and ESP32 to monitor soil moisture and automate irrigation.
- Integrated Supabase for real-time data storage and dashboards to optimize water usage and improve agricultural efficiency.

AI-Drone Threat Classifier

- Designed an AI-powered system to detect and classify drone threats using object recognition and RF signal analysis.
- Integrated blockchain for secure forensic data storage and ensured reliable security response mechanisms.

Macrowatch

- Predicts market trends (Up, Stable, Down) using Machine Learning techniques and macroeconomic indicators.
- Leverages models like Logistic Regression, Random Forest, and XGBoost with feature engineering and data preprocessing.
- Helps forecast financial market behavior and supports data-driven investment analysis.

PACTUS – Proof Authenticated Consensus Threat System

- Developed a decentralized security protocol that detects malicious agentic browsers using invisible trap codes embedded within website assets without affecting user experience.
- Records unsafe browser agents on blockchain-based smart contracts, enabling shared threat intelligence and decentralized protection across all websites using Pactus.

ParaCipher

- Built a blockchain-based parametric insurance platform enabling automated, per-shift protection for gig workers.
- Designed rule-based smart contracts to trigger instant payouts without manual claims or human discretion.
- Developed a scalable business model focused on low-cost automation, risk pooling, and platform integrations.

Achievements

- Won two bounties at DeFy'26 Hackathon for building ParaCipher, a blockchain-powered parametric insurance solution for gig workers.
- Recognized for innovative use of smart contracts to enable automated, transparent, and scalable insurance payouts.